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range-if

Abstract

This paper proposes adding a ranged if-statement, branching based on the emptiness of the supplied range. If the range is non-empty it is equivalent to range-for, otherwise an optional else part is executed.

Tony Table

Before	Proposed
<pre>// ! if in disguise! for(auto & x : a-view-pipeline std::views::take(1)) { ... }</pre>	<pre>if(auto & x : a-view-pipeline std::views::take(1)) { ... }</pre>
<pre>auto && r = a-view-pipeline; if(!r.empty()) { // ✗ not all views provide empty() for(auto & x : r) { ... } } else { //fallback for empty range }</pre>	<pre>if(auto & x : a-view-pipeline) { ... } else { //fallback for empty range }</pre>
<pre>auto && r = a-view-pipeline; // 🙄 explicit iterator use if(auto it{r.begin()}; it != r.end()) { for(; it != r.end(); ++it) { //redundant check auto & x{*it}; ... } } else { //fallback for empty range }</pre>	<pre>if(auto & x : a-view-pipeline) { ... } else { //fallback for empty range }</pre>
<pre>// 🤔 iteration flag auto empty{true}; for(auto & x : a-view-pipeline) { empty = false; ... } if(empty) { //fallback for empty range }</pre>	<pre>if(auto & x : a-view-pipeline) { ... } else { //fallback for empty range }</pre>

Revisions

R0: Initial version

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Motivation

With the adoption of range-for and especially with the introduction of ranges and views it has become possible to express C++ programs in a more declarative fashion than before, allowing regular users to avoid dealing with iterators directly. Unfortunately this programming model only supports loops, not conditionals - there is no simple way to detect an empty view and execute some alternative code path.

Readers may initially want to push back on the above assertion, pointing to the empty member function that all views inherit from `view_interface`. But said member function is actually constrained and is not available for several types of views, among them `input_views` (like generator). The only way to determine whether any given view is empty is by equality comparing its `begin-iterator` and `end-sentinel`, thereby leaving the „declarative world“ and going back to low-level constructs. As obtaining the `begin-iterator` may be non-repeatable (e.g. see the contract of `generator::begin`) naïve constructs like iterator-comparison followed by range-for are also highly problematic.

Another „workaround“ we’ve encountered in the wild was the combination of range-for and an iteration-has-taken-place flag on which a subsequent branch is taken. Neither this nor manual iterator use is a solution we want to promote to regular users. Whilst we could come up with a library-based solution, it would suffer the same limitations as `for_each` does in comparison to range-for. Therefore we prefer a dedicated language feature.

Design Space

Admittedly we weren’t too happy with re-using the `if` keyword, but remain unconvinced that the considered alternatives are superior:

- `for (init-statementopt for-range-declaration : for-range-initializer) statement1 else statement2`
works in Python with different semantics, but would change the meaning of existing C++ code.
- `if constexpropt for (init-statementopt for-range-declaration : for-range-initializer) statement1 else statement2`
expresses the wrong breaking semantics for the `else-path`.
- `for if constexpropt(init-statementopt for-range-declaration : for-range-initializer) statement1 else statement2`
looks like a conditional loop-body, but would express the right breaking semantics for the whole construct.
- Introducing a new keyword is always a hassle for existing codebases.

Our imagined syntax for range-if is a combination of the syntaxes of regular `if` and range-for, that desugars in a similar fashion to the latter:

An `if` statement of the form

```
if constexpropt(init-statementopt for-range-declaration : for-range-initializer)
statement
```

is equivalent to

```
{
    init-statementopt
    auto && range = for-range-initializer;
    auto begin = begin-expr;
    auto end = end-expr;
    if constexpropt(begin != end)
    do {
        for-range-declaration = *begin;
        statement
    } while(((void)++begin), begin != end);
}
```

and and if statement of the form

```
if constexpropt(init-statementopt for-range-declaration : for-range-initializer)  
statement1 else statement2
```

is equivalent to

```
{  
    init-statementopt  
    auto && range = for-range-initializer;  
    auto begin = begin-expr;  
    auto end = end-expr;  
    if constexpropt(begin != end)  
        do {  
            for-range-declaration = *begin;  
            statement1  
        } while(((void)++begin), begin != end);  
    else  
        do { statement2 } while(false);  
}
```

Like all other loops range-if supports break and continue. To ensure these jumps appertain to the same statement in the unprecedented else part of range-if the proposed desugaring wraps the else-statements in a dummy loop.

We are currently not aware of a need for this functionality in the context of template for, as based on our reading of P1306R5 expansions over ranges require sized ranges. If such a need ever arises, we expect template if constexpr_{opt} to be an appropriate evolutionary path.

Impact on the Standard

This proposal should be a pure language addition, the proposed syntax is unambiguous and currently invalid.

Implementation Experience

Not yet.

Proposed Wording

Wording will be provided in a future revision, if further work on this subject is encouraged.

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